

Time series trend analysis and forecasting of climate variability using deep learning in Thailand

Muhammad Waqas^{a,b}, Usa Wannasingha Humphries^{c,*}, Phyto Thandar Hlaing^{a,b}

^a The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

^b Center of Excellence on Energy Technology and Environment (CEE), Ministry of Higher Education, Science, Research and Innovation, Bangkok, Thailand

^c Department of Mathematics, Faculty of Science, King Mongkut's University of Technology Thonburi (KMUTT), Bangkok, 10140, Thailand

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ABSTRACT

Climate variability, trend analysis, and accurate forecasting are vital in a country's effective water resource management and strategic planning. Precipitation and temperature are critical indicators for assessing the effects of climate change (CC) variability. Thailand is sensitive to climatic variations, affecting the socio-economic conditions. This study quantifies climate variability and trends analysis based on precipitation, mean temperature (T_{mean}), and daily temperature range (DTR) across five climatic regions of Thailand. The results indicate regional variations: in the Central and Southern regions, there are increases in precipitation and warming temperatures, with substantial upward trends in annual precipitation (0.093 mm/year and 0.148 mm/year) and T_{mean} (0.002 °C/year). The Eastern and Northeastern regions display complex patterns with increased precipitation and temperatures. Also, DTR trends across regions show a decrease in temperature variability. The study offers new insights into forecasting climate variables for the different regions of Thailand between 2023 and 2028 by utilizing two deep learning (DL) algorithms: Wavelet-CNN-LSTM and Wavelet-LSTM, which reveals high predictive accuracy. For precipitation forecasting, Wavelet-CNN-LSTM showed higher performance in the eastern region ($R^2 = 0.83$) and comparative efficiency in other regions. Both models faced challenges in precipitation forecasting in the northeastern and southern regions. These models performed efficiently for the DTR forecast, especially in the northern region ($R^2 = 0.87$ and 0.86). For T_{mean} , both models perform similarly with high R^2 (0.57–0.87) across all regions, suggesting a substantial model accuracy. Wavelet-CNN-LSTM provides consistent performance for DTR and T_{mean} forecasting. These findings underscore the importance of climate analysis and refined forecasting models.

1. Introduction

The impact of climate change (CC) and its variability correspond to some of the most significant environmental challenges threatening the earth and humanity [1]. Global average temperature has increased by $0.74 \text{ }^{\circ}\text{C} \pm 0.18 \text{ }^{\circ}\text{C}$ during 1906–2005 [2]. This warming trend has expedited and intensified the global water cycle, resulting in severe weather events, including storms, floods, and droughts [3]. This increased intensity impacts agriculture, health, water resources, ecosystems, food security, and industrial sectors [4,5]. However, CC is inconsistent and varies critically from place to place [6,7]. Precipitation and temperature are important indicators used to monitor the extent and impact of CC variability [8,9]. Although global mean temperature and precipitation trends during interannual to millennial time intervals are

consistent across climate models and reconstructions, there is a variance in slow climate variability at regional levels. This contradiction underscores the need for more research on climatic variability [10–12]. Regional fluctuations in precipitation and temperature can be much more significant, with spatial and temporal disparities between places with different climatic conditions [13,14]. As a result, global or continental-scale climate measurements may be of little relevance for local or regional planning [15].

In the Asian continent, the southeast region is afflicted by CC [16]. With its many island countries, Southeast Asia is one of the most vulnerable regions to CC [17]. Southeast Asian countries suffer climate-related difficulties, including livelihood disruption, food shortages, disease outbreaks, and forced migration [18]. Thailand, Southeast Asia's second-largest economy, contributes to the region's development

* Corresponding author.

E-mail address: usa.wan@kmutt.ac.th (U.W. Humphries).

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[19]. Due to its geographic position, Thailand is vulnerable to CC's effects and natural climatic changes [20]. Over the last 30 years, it had an average annual temperature of 27 °C and 1587.5 mm of precipitation. However, CC has disturbed these patterns, with a 1 °C rise in average temperature and inconsistent precipitation resulting in droughts and floods [21]. Thailand ranked 9th among 178 nations most affected by CC impacts in 2021, with a 146-fold increase in natural disasters over 21 years [22]. CC impacts agricultural productivity through complex environmental interactions [19,23,24]. Many studies highlighted CC's impacts; for example, a decrease in average annual and monsoonal rainfall has been observed in northern Thailand, with predictions indicating further reductions by 2030. This region experiences increased drought risks caused by CC [25]. Northeastern Thailand, a major agricultural area, faces recurring droughts every six years, severely impacting crop yields [26]. In southern Thailand, extreme precipitation trends show rising intensity and declining wet days, while interannual variability reveals that December–January–February have more variability than June–July–August [27]. The southern region's rainfall projections suggest increased precipitation under future scenarios, impacting flood risk and water management [28]. Therefore, there is a need to study the climate variability and past trends for Thailand overall. Precipitation and temperature are crucial to understanding climate variability [29–31]. In today's evolving climate, examining potential shifts in key climatic factors within this vulnerable ecosystem is crucial. This study focuses on two fundamental climatic parameters: temperature and precipitation. Regional-scale analyses offer valuable insights that inform the management and planning of local economic and socio-cultural activities. Understanding historical trends in these variables is a critical step before developing accurate predictive models for future climate scenarios. To study climate variability, many studies considered the trends and patterns in precipitation and temperature across various regions globally [32].

In Thailand, the Mann-Kendall (MK) test and Sen's slope estimator (SSE) are utilized to analyze rainfall trends in the Mae Klong River Basin from 1971 to 2015, revealing increases in annual rainfall at 75 % of monitoring stations and in the dry season at 5 out of 8 stations in the upper basin [33]. For temperature and humidity trends across Thailand, the MK test and new trend analysis methods indicated an increasing trend in temperature and humidex, with increases in the southern and eastern regions [34]. Similarly, hydrological trends were assessed via the MK test, and varied trends in precipitation, inflow, outflow, and storage for five major dams were observed [35]. The Diurnal Temperature Range (DTR) analysis using MK and SSE methods indicated increasing temperatures with fluctuating DTR trends [36]. In the Ping Basin, historical data and GCM projections revealed increases in minimum temperature and variable precipitation trends [37]. In another study, the MK and SSE applied to extreme climate indices in North Thailand demonstrated temperature extremes and a slight increase in annual rainfall [38]. The literature demonstrates that the MK and SSE methods have been applied to identify trends across various regions and river basins. Based on the literature, the MK and SSE methods were selected for this study to analyze and determine past trends in precipitation, T_{mean} , and DTR for each region of Thailand.

Together with trend analysis, it is crucial to forecast this trend for the future.

Weather forecasting models are divided into model-driven models (MDMs) and data-driven models (DDMs). MDMs focus on physics-based atmospheric processes, while the increasing complexity of spatio-temporal meteorological data has led to the rise of DDMs, including AI and statistical methods. DDMs address the limitations of MDMs by leveraging advanced deep learning (DL) and machine learning (ML) techniques to uncover complex patterns [39]. DL and ML have transformed forecasting, but it has some limitations. One limitation is the weak interpretability of artificial intelligence models, often criticized as "black boxes" due to their lack of transparency in decision-making processes [20,40]. Despite limitations, with improving data quality

and model interpretability, many studies have forecasted short- and long-term climate variability and other variables using DL and ML models to provide information about climate impact [20,25,28,39,41–44]. For example, in central Thailand, the hybrid technique combined biorthogonal wavelet transformation with Radial Basis Function Neural Networks and Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) showed superior performance over conventional models in precipitation prediction [39]. LSTM and Genetic Programming effectively downscaled monthly rainfall for the Thale Sap Songkhla basin and outperformed other ML techniques [44,45]. In the Mekong River basin, LSTM models provided more accurate runoff predictions than the Soil and Water Assessment Tool (SWAT), indicating their efficacy for hydrologic modeling [46]. Furthermore, CNN and LSTM models predicted streamflow changes and flood risks in the 3S River Basin under various climate scenarios [42].

This study is crucial due to the impacts of climate variability on Thailand. The study uses DL algorithms and time series trend analysis to produce accurate long-term projections of climate trends that are essential for resource management and adaptation. The study has two main goals: (1) to assess climate variability and trends across five distinct climatic regions in Thailand through SSE and the MK test for monthly, seasonal, and annual temperature and precipitation, which are essential for effective water resource management; and (2) to provide advanced forecasting insights using hybrid DL models.

2. Materials and methods

2.1. Study area

Thailand is located between the latitudes of 5°37' and 20°27' N and longitudes of 97°22' and 105°37' E (Fig. 1) [47]. Based on climate patterns and meteorological conditions, Thailand can be categorized into five distinct climatic regions, each characterized by unique topographical features, such as (a) central, (b) eastern, (c) northeastern, (d) northern, and (e) southern [48]. Thailand has a tropical climate characterized by continuous warmth throughout the year [49,50]. The rainy season lasts from mid-May to mid-October and is driven by the southwest monsoon, which brings warm, humid air from the Indian Ocean, resulting in heavy rainfall. July and August are often the rainiest months. The northeast monsoon influences the winter season, which lasts from mid-October to mid-February. It brings colder, drier air from China to northern Thailand. Summer, which lasts from mid-February to mid-May, is a transitional period defined by increasing temperatures as the region switches from the Northeast to the southwest monsoon, with April being the hottest month [51,52].

2.2. Data collection and quality checking

The initial data acquisition phase focused on defining the temporal scope and identifying key variables essential for the study. The availability of observational datasets across Thailand's geographical extent was assessed to ensure the inclusion of reliable meteorological data. A comprehensive evaluation of meteorological records from the TMD was conducted, revealing variations in data availability across different stations, as shown in Fig. S1 (Supplementary File). The TMD's network of stations, established in 1961, has expanded, with newer stations installed after 2018. Ensuring uniform data availability across all stations for the chosen variables proved challenging, and after careful consideration of data completeness, 30 years from 1993 to 2022 were selected for the study. Daily data of precipitation and minimum and maximum temperature datasets were collected from selected 129 TMD stations for 30 years. Many stations contain missing data during this duration, as shown in Figs. S2–S4. For handling missing data, this study employed the LSTM-RNN approach developed by Wangwongchai et al. (2023) [53] for the estimation of missing values in the precipitation and temperature dataset, as shown in Fig. 2 (Stage I). The stations with more

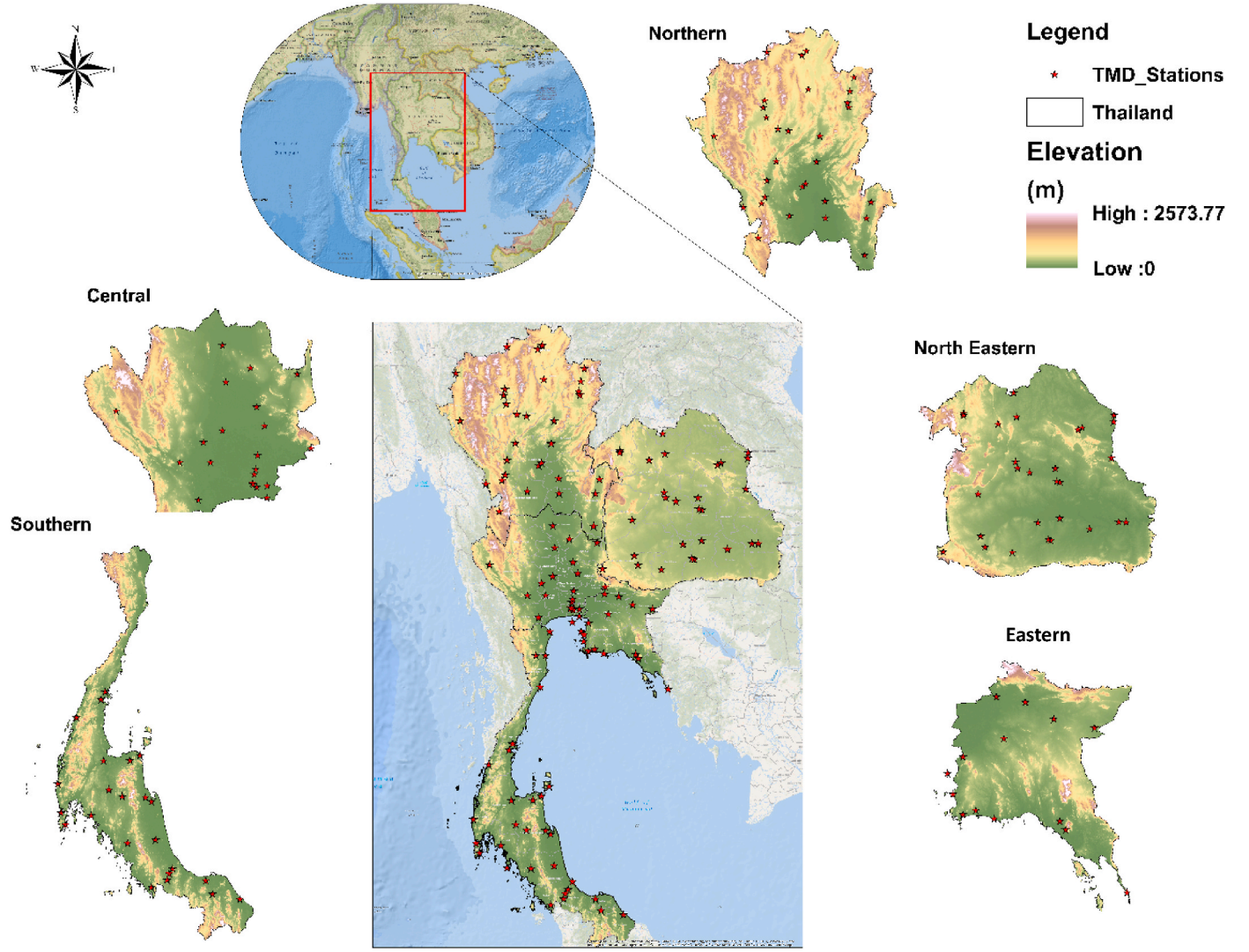


Fig. 1. TMD station's distribution over different regions of Thailand.

than 50 % missing values are skipped from the total stations. The potential for recurring observational or processing missing data within a hydrological year required thorough checks to maintain data consistency. To detect significant deviations from normal data ranges, the study utilized the Grubbs and Beck (1972) method [54]. A thorough quality assessment and control measures were implemented to maintain data accuracy. These measures also included removing outliers and addressing continuity issues [53]. These procedures were essential to ensure the reliability of the meteorological indices utilized in the analysis. Following a thorough quality control process, each station's monthly data were organized by region. Subsequently, T_{mean} and DTR were systematically computed for each dataset, ensuring accurate and region-specific climate trend assessments.

2.3. Trend analysis methods

2.3.1. Mann-Kendall test

The Mann-Kendall (MK) test [55] is a widely applied non-parametric statistical method for analyzing trends in time series data, particularly in fields like climatology and hydrology [36,56]. MK test was introduced in 1945 and has since gained global recognition [55]. Its key advantages include not requiring distributed data and being relatively insensitive to immediate changes in heterogeneous time series [57]. The MK test statistic, denoted as S , is calculated using the following formula:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(x_j - x_i) \quad (1)$$

$$\text{sgn}(x_j - x_i) = \begin{cases} 1 & \text{if } x_j - x_i > 1 \\ 0 & \text{if } x_j - x_i = 1 \\ -1 & \text{if } x_j - x_i < 1 \end{cases} \quad (2)$$

x_i and x_j represent the values at times i and j , respectively, and n denotes the dataset's length. Both the positive and negative scores of the S represent the dataset's increasing and decreasing tendencies. The following equation in this study is rejected when the dataset (n) length surpasses 10.

$$\text{VAR}(S) = \frac{n(n-1)(2n+5) - \sum_{i=1}^n t_i(i-1)(2i+5)}{18} \quad (3)$$

t_i denotes the total amount of data values. The Z-value is derived after finding the variability of the time data series using the following equation:

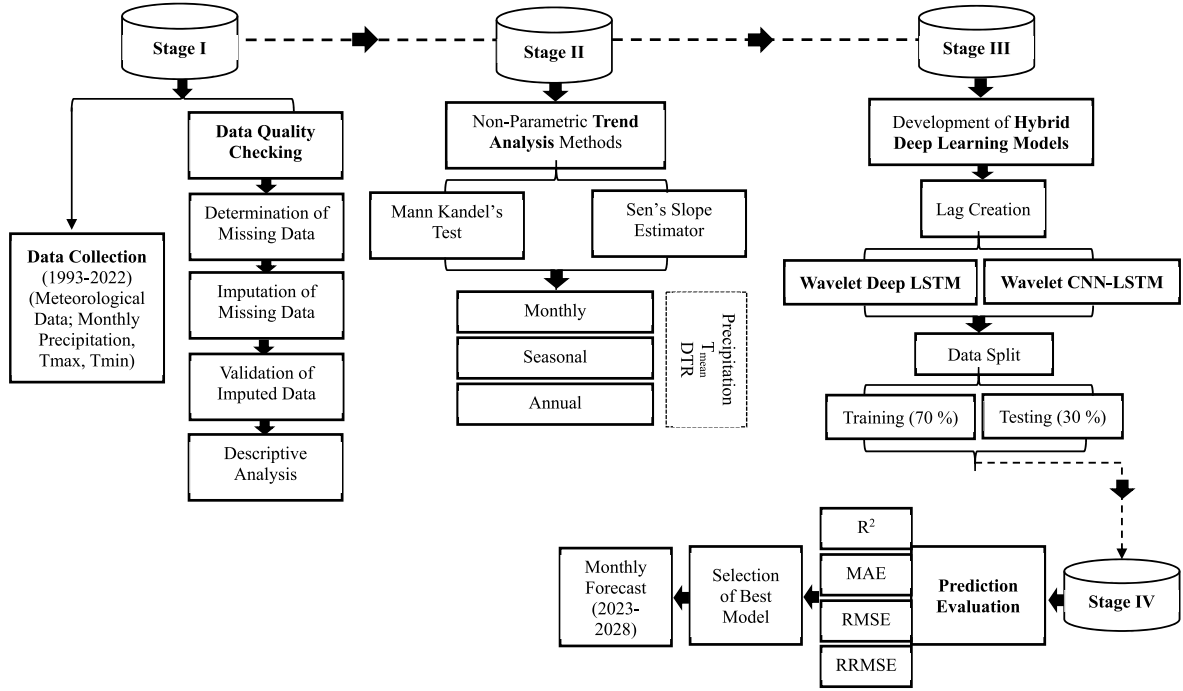


Fig. 2. Overall methodology used for time series trend analysis and forecasting by deep learning models.

$$Z = \frac{S - 1}{\sqrt{\text{var}(S)}} \quad \text{if } S > 0$$

$$Z = 0 \quad \text{if } S = 0$$

$$Z = \frac{S + 1}{\sqrt{\text{var}(S)}} \quad \text{if } S < 0$$
(4)

($\alpha = 1\%$, $\alpha = 5\%$, and $\alpha = 10\%$) normal distribution ranges with significance levels (α) are then compared to the Z [58]. The significance levels $\alpha = 1\%$, $\alpha = 5\%$, and $\alpha = 10\%$ were chosen to provide varying degrees of confidence in detecting trends using the MK test. At $\alpha = 1\%$, a highly significant trend is identified, reducing false positives. At $\alpha = 5\%$, the commonly accepted threshold balances sensitivity and reliability, while $\alpha = 10\%$ allows for detecting weaker, emerging trends. These levels ensure robust trend analysis by accommodating strong and subtle trends in the time series data.

The inclusion of these significance levels allows for a broader interpretation of trends in precipitation, T_{mean} , and DTR. For instance, at the 1% level, only the strongest trends are considered, while at the 10% level, weaker trends can also be identified. This multi-tiered approach enhances the robustness of the analysis by providing insight into the confidence level required to observe different magnitudes of trends within the data.

2.3.2. Sen's slope estimation

The Sen's slope (β) estimator is a non-parametric method used to measure the magnitude and rate of change in time series data [59]. One of its key advantages is its ability to handle data with outliers. The slope (β) is calculated by taking the median of the slopes between all pairs of points in the time series. Sen introduced this method in 1968 [60].

For the determination of the true slope of a trend, expressed as a change per year, Sen's Slope estimator [61] is employed. This method is applicable when the trend is assumed to be linear, implying that $f(t)$ in equation (5) is linear [56].

Mathematically:

$$f(t) = Xt + \beta$$
(5)

B represents a constant, and X is the slope.

To determine X_t , it is necessary to compute the slope for all pairs of data points in the dataset.

$$X_i = \frac{x_j - x_k}{j - k}$$
(6)

If $j > k$, slope estimates X_i if in x_j there are n values in the time series, we will get $N = \frac{n(n-1)}{2}$. The SSE is the median of N values, and these values are ranked from smallest to largest.

Mathematically, Sen's estimator is:

$$X = X_{[n(n-1)/2]} \quad \text{if } n \text{ is odd}$$
(7)

$$X = \frac{1}{2} (X_{[n/2]} + X_{[(n+2)/2]}) \quad \text{if } n \text{ is even}$$
(8)

The non-parametric technique on the normal distribution provides a two-sided confidence range of 100 (1- α) % for estimating the slope. It is valid if n is more than 10.

In this study, the confidence level is calculated at two distinct levels. 1) $\alpha = 0.01$ and 2) $\alpha = 0.05$. The confidence levels $\alpha = 0.01$ and $\alpha = 0.05$ were selected to provide a range of significance for detecting trends using SSE. The $\alpha = 0.01$ level ensures strong confidence in rejecting the null hypothesis, while $\alpha = 0.05$ is a commonly used threshold for balancing sensitivity and reliability in trend analysis. These levels help capture both strong and moderate trends in the data.

To get these confidence levels, first, we compute,

$$C_\alpha = Z_{1-\alpha/2} \sqrt{\text{VAR}(S)}$$
(9)

$Z_{(1-\alpha/2)}$ is derived from the normal distribution, whereas $\text{VAR}(S)$ is already defined in equation (3). $M_1 = (N - C_\alpha)/2$ is calculated [59].

2.4. Forecasting deep learning models

2.4.1. Wavelet decomposed CNN-LSTM

DL has advanced from near-human to super-human performance in tasks like voice-to-text translation [62], object detection [63], and forecasting hydrological components [39]. This era, marked by the work of Hinton and Salakhutdinov in 2006 [64], highlighted the importance

of depth in ANNs. This development has impacted various applications. It shows that ANNs with several hidden layers have superior learning capabilities, which can be boosted further by increasing their depth. It gave rise to the term "DL," a subfield of ML that excels at dealing with complicated patterns and massive datasets [20]. This section explores forecasting models' fundamental principles and development, emphasizing wavelet-decomposed LSTM and deep CNNs. It will briefly examine 1D CNNs designed to process one-dimensional signals and data sets. Special attention will be given to the compact and adaptive 1D CNN models that integrate wavelet transforms with LSTM architectures. The accuracy of LSTM models can be constrained by the length of the input sequence, a limitation known as the vanishing gradient problem, which may lead to the loss of crucial historical information [65]. The CNN-LSTM architecture is employed to overcome this, with CNN layers for feature abstraction and LSTM for sequence prediction. This model is advantageous as it reduces the input sequence length by extracting key temporal features through convolutional filters [66], allowing the LSTM layer to more effectively capture temporal dependencies [67]. Fourier Transform (FT) has traditionally been the standard technique for spectral decomposition in input signal processing [68]. However, FT's basis function spans the entire time domain, leading to a loss of temporal frequency information. To address this limitation, the Short-Time Fourier Transform (STFT) was introduced [69]. STFT applies a fixed time window to segment the signal before performing the Fourier transform, allowing the capture of both time and frequency domain information.

Nevertheless, selecting an appropriate window width remains challenging, as STFT's time and frequency resolutions are fixed and cannot be dynamically adjusted based on frequency [70]. This study utilized Wavelet Transformation (WT), incorporating both Continuous Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT). The CWT calculates the product of the signal and the wavelet function, resulting in a continuous and comprehensive time-frequency representation. Although CWT provides an in-depth analysis, the wavelet coefficients it generates can sometimes be redundant [47]. The DWT mitigates redundancy by employing discretely sampled wavelets, thereby decreasing the volume of data necessary for processing [71]. DWT is a powerful tool for analyzing signal patterns in both the time and frequency domains [72]. DWT effectively filters out noise and captures essential timescale information in complex signals, such as non-linear and non-stationary precipitation rates [73,74]. The bior1.3 wavelet at level 2 was selected after evaluating various wavelet families and decomposition levels, as it provided the best balance between noise reduction and the capture of essential timescale information [75,76]. This wavelet's biorthogonal nature ensures smooth reconstruction and detailed signal analysis, making it ideal for identifying local temporal patterns in non-linear, non-stationary precipitation and temperature data. Testing multiple wavelet options and levels minimized sensitivity to these choices to ensure robust results.

In the next stage, the decomposed input signal is processed to 1D-CNN. CNNs, an extension of Multilayer Perceptrons (MLPs) within DL, are designed for effective data extraction and processing through convolution. It made them suitable for classification, identification, and prediction [77]. CNNs consist of two primary layers: feature extraction layers, convolution and subsampling, and Fully Connected layers for classification or prediction [78]. A 1D CNN is a specialized form of CNN where convolution and pooling operations are performed in one dimension, well-suited for sequential or unidirectional data processing, such as time series weather data [79]. The architecture of CNNs must be tailored to the specific input data type, as 2D CNNs are unsuitable for processing 1D weather data due to the dimensional mismatch in convolution and pooling operations [80]. Critical parameters in the convolution layer, such as feature maps, filter sizes, activation functions, stride, and padding, are crucial for CNN performance [78]. In this study, input vectors undergo convolution with kernel filters during training, followed by ReLU activation to eliminate negative values and

max-pooling to reduce layer size, thereby optimizing information extraction in the convolutional layers and other nodes and parameters shown in Fig. 3.

The Wavelet-LSTM and Wavelet-CNN-LSTM models were applied for forecasting precipitation, T_{mean} , and DTR for 2023–2028. The performance of both models was evaluated using several metrics, including the Coefficient of Determination (R^2), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Relative RMSE, and Mean Absolute Percentage Error (MAPE). A comprehensive explanation of these evaluation criteria is provided in the Supplementary File. This study was implemented in the Python 3.11 environment, using TensorFlow and other relevant libraries to serve as the deep learning framework for model training, prediction, and comparative analysis.

2.4.2. Wavelet LSTM

The second hybrid model integrates wavelet decomposition with LSTM networks proposed by Waqas et al. (2024) to enhance forecasting [76]. In Wavelet LSTM, the model is developed in three phases: wavelet decomposition, model construction, training, validation, and testing. Initially, wavelet transformation is applied to decompose the input data into wavelets, providing a detailed time-frequency representation [81]. In the model construction phase, the LSTM network utilizes an internal state and three key gates—forget, input, and output—to manage information flow and update the cell state, influencing the forecasted output, as shown in Fig. S5. The architecture employs two LSTM layers with 50 hidden nodes, leveraging sigmoid and tanh activation functions to process and retain crucial information. The chosen mother wavelet function $\psi(t)$, specifically the Bior1.3 wavelet at level 2, functions as a sequence of configurable moving windows for local temporal pattern identification [75,76]. In the second stage, the LSTM model contains an internal state and three important gates within each LSTM cell: forget, input, and output [82].

3. Results and discussion

3.1. Time series trend analysis for the central region of Thailand

The time series trend analysis of precipitation, T_{mean} , and DTR for the Central region of Thailand from 1993 to 2022 is summarized in Table 1, Table 2, and Table 3, respectively. The analysis of precipitation reveals temporal variability and trend patterns. In most months, the MK test results indicate trends, particularly in February, June, July, August, and September, where increasing trends in precipitation are observed with p -values < 0.05 . This suggests a strong upward trend in precipitation during these months. The SSE identified the highest slopes in August and September, with values of 2.017 mm/year and 2.448 mm/year, respectively, indicating a substantial increase in rainfall during the rainy season.

Conversely, March, October, and December exhibit non-significant trends, implying that precipitation in these months has remained relatively stable. Seasonal and annual trends further show a marked increase in precipitation. The rainy season displays the highest significance level (p -value < 0.001), with a Sen's slope of 0.166 mm/year, suggesting a pronounced intensification of rainfall during the monsoon period. Similarly, annual precipitation shows an upward trend with a p -value < 0.001 and a Sen's slope of 0.093 mm/year, reflecting a steady increase in total precipitation over the last three decades.

For T_{mean} , the results highlight warming trends across several months. March, June, July, August, September, October, and November all show positive trends, with p -values < 0.05 , providing robust evidence of ongoing warming. Notably, March, August, and September exhibit high trends with p -values < 0.01 and Sen's slopes of 0.032 °C/year, 0.024 °C/year, and 0.025 °C/year, respectively, indicating substantial increases in mean temperatures during these months. Seasonal trends also support this finding, with warming observed in the summer and rainy seasons. The annual mean temperature demonstrates a significant

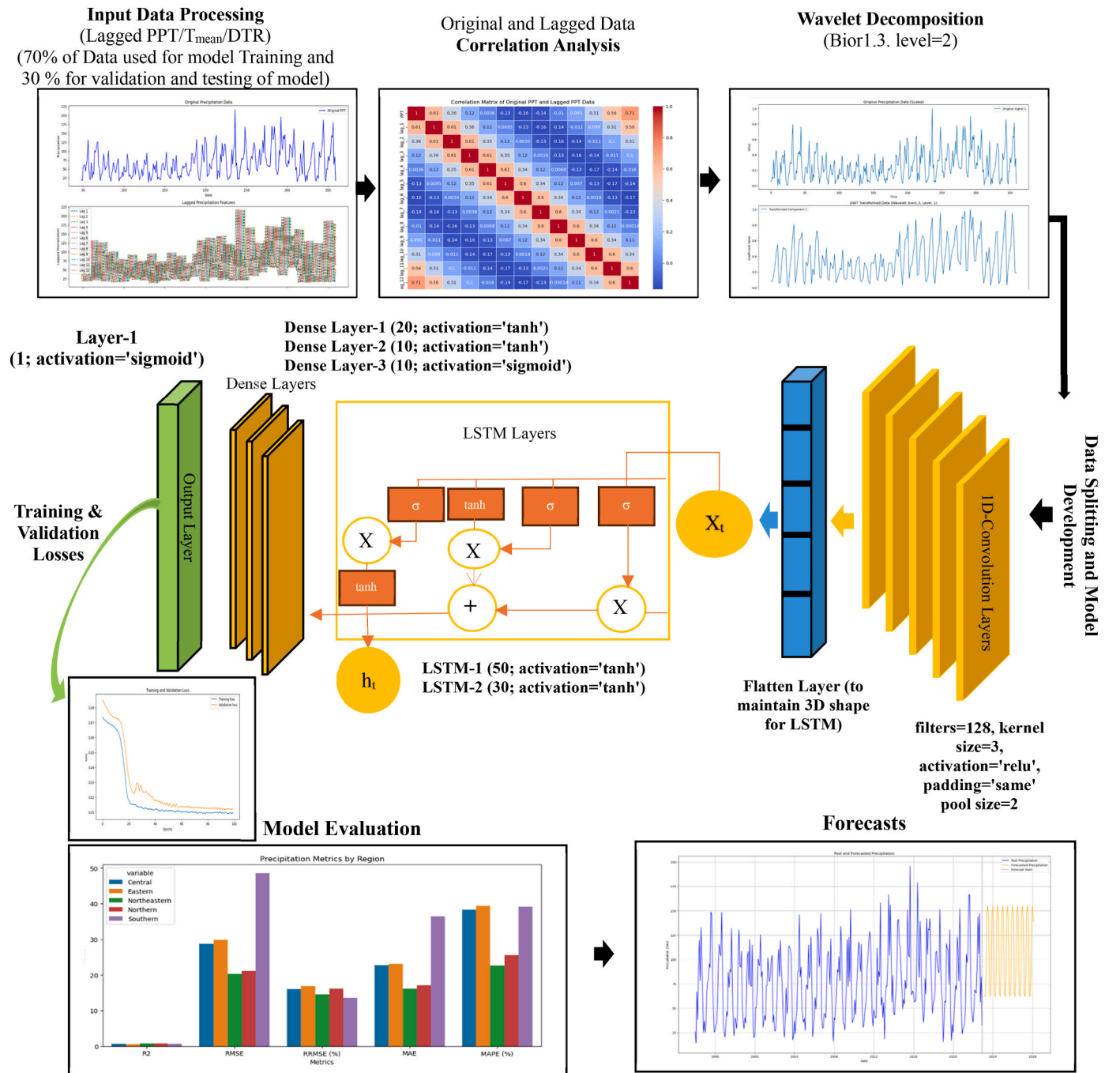


Fig. 3. The working mechanism of wavelet decomposed CNN-LSTM forecasting model.

upward trend with a Sen's slope of $0.002\text{ }^{\circ}\text{C}/\text{year}$, pointing to a gradual, long-term warming trend in the Central region. While the rate of increase might appear small, its cumulative impact over the 30-year study period is notable, translating into a $0.06\text{ }^{\circ}\text{C}$ rise. This continuous warming could exacerbate heat stress, affecting agricultural productivity and human health.

The analysis of DTR presents a more complex picture. DTR shows mostly non-significant trends, with most months having p-values greater than 0.05. However, although these are not statistically significant, May, June, and July display slight positive trends in DTR. The annual DTR exhibits a negative trend, with a p-value of 0.106 and a Sen's slope of $-0.001\text{ }^{\circ}\text{C}/\text{year}$, suggesting a slight decrease in the DTR over time. While minor, this reduction in DTR may indicate narrowing temperature ranges between day and night, which could have implications for

temperature-sensitive crops.

The threshold for statistical significance is consistently set at a p-value <0.05 for strong evidence, with marginal significance indicated for p-values between 0.05 and 0.10. Additionally, the practical significance of trends has been clarified. For example, a Sen's slope of $0.002\text{ }^{\circ}\text{C}/\text{year}$ in T_{mean} represents a cumulative increase of $0.06\text{ }^{\circ}\text{C}$ over 30 years, which, though gradual, could lead to meaningful real-world impacts. The observed changes in seasonal trends, particularly the intensification of rainfall during the monsoon, may signal shifts in traditional patterns, potentially changing Thailand's agricultural cycles and water management strategies.

3.1.1. Time series trend analysis for the eastern region of Thailand

The analysis of precipitation data for the Eastern region reveals

Table 1

MK and SSE results for Precipitation for Central Region.

Month	Min (mm)	Max (mm)	Mean (mm)	SD (mm)	CV (%)	MK Z value	MK P value	Sen's slope (mm/year)	Trend/Significance
January	13.751	68.413	31.945	14.822	46.398	1.748	0.080	0.532	▲*
February	15.375	77.221	35.184	17.421	49.514	2.640	0.008	0.737	▲***
March	23.093	105.662	51.839	21.438	41.355	0.928	0.354	0.369	▲
April	29.637	119.485	59.294	24.321	41.017	2.105	0.035	1.017	▲**
May	43.901	196.795	85.922	28.943	33.685	1.963	0.050	0.926	▲**
June	31.885	170.555	76.974	28.162	36.586	3.104	0.002	1.551	▲***
July	34.407	143.810	79.906	30.053	37.611	2.640	0.008	1.882	▲***
August	40.065	144.089	94.216	32.582	34.582	2.640	0.008	2.017	▲***
September	68.231	217.017	126.117	40.628	32.215	2.319	0.020	2.448	▲**
October	21.443	145.961	81.352	34.381	42.262	1.570	0.116	1.235	▲
November	12.684	94.916	39.994	21.841	54.612	2.034	0.042	0.980	▲**
December	15.620	60.625	31.824	12.734	40.013	1.534	0.125	0.572	▲
Winter	13.751	77.221	32.984	15.025	45.553	3.374	0.001	0.048	▲***
Summer	23.093	196.795	65.685	28.843	43.911	2.656	0.008	0.075	▲***
Rainy	31.885	217.017	94.303	38.177	40.483	5.076	0.000	0.166	▲***
Annual	12.684	217.017	66.214	38.686	58.425	5.015	0.000	0.093	▲***

Table 2

MK and SSE results for T_{mean} for Central Region.

Month	Min (°C)	Max (°C)	Mean (°C)	SD (°C)	CV (%)	MK Z value	MK P value	Sen's slope (°C/year)	Trend/Significance
January	24.267	28.680	26.665	0.998	3.741	1.142	0.254	0.029	▲
February	26.784	29.625	28.251	0.824	2.918	0.963	0.335	0.017	▲
March	27.196	31.039	29.899	0.731	2.444	2.748	0.006	0.032	▲***
April	29.617	32.756	30.901	0.756	2.448	0.642	0.521	0.011	▲
May	28.860	32.264	30.478	0.927	3.043	1.677	0.094	0.037	▲*
June	29.032	30.791	29.893	0.533	1.784	2.462	0.014	0.033	▲**
July	28.620	30.398	29.382	0.428	1.456	2.141	0.032	0.018	▲**
August	28.428	29.993	29.190	0.368	1.260	2.926	0.003	0.024	▲***
September	28.298	29.766	28.873	0.365	1.262	3.033	0.002	0.025	▲***
October	27.394	29.725	28.510	0.471	1.653	2.248	0.025	0.022	▲**
November	25.927	29.309	27.824	0.800	2.877	2.890	0.004	0.045	▲***
December	23.560	28.365	26.316	1.188	4.513	1.106	0.269	0.036	▲
Winter	23.560	29.625	27.077	1.312	4.847	1.331	0.183	0.002	▲
Summer	27.196	32.756	30.426	0.901	2.960	2.635	0.008	0.002	▲***
Rainy	28.298	30.791	29.335	0.563	1.921	3.908	0.000	0.002	▲***
Annual	23.560	32.756	28.849	1.546	5.358	2.475	0.013	0.002	▲**

Table 3

MK and SSE results for DTR for Central Region.

Month	Min (°C)	Max (°C)	Mean (°C)	SD (°C)	CV (%)	MK Z value	MK P value	Sen's slope (°C/year)	Trend/Significance
January	8.585	12.426	10.880	0.924	8.497	-1.820	0.069	-0.045	▼*
February	9.186	12.016	10.658	0.721	6.763	-1.748	0.080	-0.035	▼*
March	7.976	11.262	10.078	0.667	6.620	-1.784	0.074	-0.017	▼*
April	8.667	10.820	9.930	0.529	5.328	-0.535	0.592	-0.006	▼
May	7.702	9.882	8.922	0.643	7.210	0.357	0.721	0.005	▼
June	7.294	9.794	8.305	0.584	7.031	0.678	0.498	0.008	▲
July	6.930	8.913	7.900	0.445	5.639	0.071	0.943	0.000	▲
August	7.196	8.493	7.852	0.343	4.363	0.143	0.887	0.002	▲
September	7.239	8.804	7.907	0.412	5.206	-0.214	0.830	-0.003	▼
October	6.594	9.081	7.944	0.580	7.300	-0.535	0.592	-0.004	▼
November	7.758	10.236	8.898	0.747	8.392	-1.035	0.301	-0.019	▼
December	8.544	12.445	10.061	0.867	8.615	-1.641	0.101	-0.033	▼
Winter	8.544	12.445	10.533	0.902	8.563	-3.018	0.003	-0.003	▼
Summer	7.702	11.262	9.643	0.799	8.284	-1.101	0.271	-0.001	▼
Rainy	6.930	9.794	7.991	0.485	6.065	0.075	0.940	0.000	▲
Annual	6.594	12.445	9.111	1.275	13.990	-1.619	0.106	-0.001	▼

notable trends across various months (Tables S1–S3). Significant increases in precipitation are observed in February, July, September, November, and December, as demonstrated by p-values <0.05. Particularly, July and September exhibit substantial increases in rainfall, with Sen's slopes of 1.287 mm/year and 1.699 mm/year, respectively. In contrast, March, April, May, June, August, and October show non-significant trends, indicating relatively stable precipitation levels during these months. Seasonal analysis highlights pronounced increases in precipitation during the rainy and winter seasons, with p-values <0.001,

and the rainy season shows a Sen's slope of 0.067 mm/year, signifying more rainfall during the monsoon. Annual precipitation trends also show an increase, with a p-value of 0.003 and a Sen's slope of 0.055 mm/year, reflecting a consistent rise in total precipitation over the past three decades.

The analysis of T_{mean} indicates warming trends in March, June, August, September, October, and November, with p-values <0.05 providing strong evidence of temperature increases. August and September display the highest significance levels, with p-values<0.01

and Sen's slopes of 0.023 °C/year and 0.021 °C/year, respectively. Seasonal trends reveal warming in the rainy and summer seasons, with p-values = 0.001 for both. The annual T_{mean} shows an upward trend (p-value = 0.016), with a Sen's slope of 0.001 °C/year, indicating a gradual long-term warming trend in the Eastern region.

The DTR analysis reveals a general decline, with negative trends in January, March, August, November, and December, as indicated by p-values <0.05. The most notable DTR reductions are in March (−0.031 °C/year) and August (−0.011 °C/year). Seasonal analysis also shows reductions in DTR during the winter and rainy seasons. The annual DTR exhibits a negative trend (p-value = 0.016) with a Sen's slope of −0.002 °C/year, indicating a gradual decrease in temperature variability. These results indicate increasing precipitation, T_{mean} , and a decline in DTR in eastern Thailand.

3.1.2. Time series trend analysis for the northeastern region of Thailand

In the Northeastern region of Thailand, the precipitation, T_{mean} , and DTR analysis reveal intricate climate variability patterns (Tables S4–S6). For precipitation, variations are evident across different months and seasons. Notably, June shows a decrease in precipitation, with a p-value of 0.101 and a Sen's slope of −1.709 mm/year, which diverges from the typical expectation of increased rainfall during the monsoon period. Conversely, July and September exhibit substantial increases in precipitation, with Sen's slopes of 3.035 mm/year and 2.390 mm/year and p-values of 0.017 and 0.042, respectively. Despite these monthly trends, the annual precipitation trend demonstrates a modest positive change but remains statistically insignificant (p-value = 0.141 and Sen's slope of 0.043 mm/year). It suggests that while there are notable variations within individual months, the overall annual trend is less pronounced.

The T_{mean} trends in the Northeastern region indicate warming during the summer and rainy seasons. Significant increases are observed in June, July, August, September, and November, with p-values of 0.005, 0.038, 0.001, 0.001, and 0.003, respectively, and Sen's slopes ranging from 0.021 °C/year to 0.058 °C/year. These findings highlight a pronounced warming trend during the warmer months. The annual T_{mean} trend also shows significance, with a p-value of 0.038 and a Sen's slope of 0.002 °C/year, reflecting a general warming pattern over the year. The seasonal trends support this observation, with warming detected in the summer and rainy seasons (p-values of 0.049 and 0.000, respectively).

Regarding DTR, a consistent decrease is observed in January, December, and the winter season. Negative trends are evident with p-values of 0.025, 0.032, and 0.004, respectively, and negative Sen's slopes ranging from −0.049 °C/year to −0.003 °C/year. This reduction in DTR suggests a narrowing temperature difference between day and night, potentially indicating a rise in minimum temperatures or decreased variability in temperature ranges. Other months, such as March, April, and October, display stable DTR without significant trends. Overall, the Northeastern region experiences a complex interplay of increased precipitation in certain months during the summer and rainy seasons, alongside warming trends and a decline in the diurnal temperature range throughout the year.

3.1.3. Time series trend analysis for the northern region

In the Northern region of Thailand, precipitation trend analysis reveals notable changes across various months and seasons (Tables S7–S9). Specifically, upward trends are observed in February and May, with p-values of 0.087 and 0.054, respectively. The Sen's slopes indicate increases in rainfall of approximately 1.285 mm/year for February and 1.815 mm/year for May. November also shows a substantial positive trend, with a p-value of 0.011 and a Sen's slope of 1.509 mm/year, reflecting increased rainfall during the late monsoon period. Seasonal and annual precipitation trends further highlight increases, with a p-value of 0.041 for summer precipitation, 0.025 for annual precipitation, and Sen's slopes of 0.096 mm/year and 0.069 mm/year, respectively. However, precipitation during the winter and rainy

seasons does not show trends, indicating that while there are variations within specific periods, the overall monsoon trend remains relatively stable.

T_{mean} reveals warming across several months, including March, July, August, September, and November, with p-values less than 0.05. Notably, March, July, and August exhibit higher Sen's slopes, with values of 0.049 °C/year, 0.023 °C/year, and 0.030 °C/year, respectively. Conversely, April and December show less pronounced warming with MK p-values of 0.695 and 0.116, indicating relatively stable temperatures during these months. Seasonal and annual temperature trends also demonstrate warming in the summer and rainy seasons, with p-values below 0.05. The Sen's slopes for these seasons are 0.003 °C/year for summer and 0.002 °C/year for the rainy season. The annual T_{mean} trend is also significant, with an MK p-value of 0.002 and a Sen's slope of 0.003 °C/year, indicating a clear upward temperature trend over the year.

Analysis of the DTR reveals a consistent decline across various months and seasons. January experiences a decrease in DTR, with a p-value of 0.011 and a Sen's slope of −0.081 °C/year, suggesting reduced temperature variability. February and August also show decreases in DTR, with p-values of 0.064 and 0.212, respectively. Seasonal and annual DTR trends indicate declines, with p-values below 0.05 for winter and annual DTR and Sen's slopes of −0.005 °C/year and −0.001 °C/year, respectively. This reduction in DTR reflects a decrease in daily temperature variability. Overall, the Northern region of Thailand exhibits increased precipitation during certain months in the rainy season, alongside warming across multiple periods and a consistent decrease in DTR.

3.1.4. Time series trend analysis for the southern region

In the Southern region of Thailand, the time series trend analysis of precipitation, T_{mean} , and DTR from 1993 to 2022 reveals several key patterns detailed in Table S10, Table S11, and Table S12, respectively. Precipitation data for the Southern region of Thailand reveal increases in February, May, and June. February exhibits a strong positive trend, with a p-value of 0.003 and a Sen's slope of 3.361 mm/year. May and June also show substantial increases, with p-values of 0.010 and 0.038 and Sen's slopes of 2.590 mm/year and 1.983 mm/year, respectively. These findings indicate a pronounced rise in rainfall during these months. The annual precipitation trend is notable, with a p-value of 0.000 and a Sen's slope of 0.148 mm/year, reflecting an overall increase in rainfall. Seasonal precipitation trends similarly demonstrate positive changes, with increases observed in the winter and rainy seasons, characterized by p-values of 0.001 and 0.006 and Sen's slopes of 0.211 mm/year and 0.110 mm/year, respectively. T_{mean} trends in the Southern region show warming across several months. Notable increases are observed in March, July, August, September, October, and November, with p-values below 0.05. March, July, August, September, and October display the most pronounced warming trends, with Sen's slopes of 0.030 °C/year, 0.019 °C/year, 0.022 °C/year, 0.019 °C/year, and 0.024 °C/year, respectively. The annual temperature trend has a p-value of 0.001 and a Sen's slope of 0.002 °C/year, indicating a steady rise in temperatures throughout the year. Seasonal temperature trends align with this warming pattern, particularly in the summer and rainy seasons, with p-values of 0.002 and 0.000 and Sen's slopes of 0.002 °C/year for both seasons. The analysis of DTR indicates a decreasing trend in the Southern region. January shows a decrease in DTR, with a p-value of 0.069 and a Sen's slope of −0.030 °C/year. The overall annual trend also reflects a decline in DTR, though less pronounced, with an MK p-value of 0.255 and a Sen's slope of −0.001 °C/year. Seasonal trends further emphasize this reduction, with declines observed in winter and summer DTR, albeit less important than the monthly trends. The Southern region trends reflect ongoing climatic shifts with potential implications for local weather patterns and ecological systems.

3.1.5. Regional comparison of trends climate variability

The time series analysis of precipitation, T_{mean} , and DTR from 1993 to 2022 across Thailand's five regions reveals climate variability. In the Central region, monthly precipitation increases during the rainy season (MK p-value <0.001), with a Sen's slope of 0.093 mm/year indicating increased rainfall intensity. T_{mean} trends exhibit warming (MK p-values <0.01) across several months, with annual mean temperatures rising. DTR trends are largely non-significant, although the annual DTR shows a slight decline. The Eastern region also experiences precipitation increases in July and September. The annual precipitation trend is upward (MK p-value = 0.003), with a Sen's slope of 0.067 mm/year. T_{mean} displays warming in August and September, while DTR decreases in multiple months, reflecting reduced diurnal variability. In the Northeast, precipitation increases occur in July and September, but the overall annual trend is non-significant. T_{mean} trends indicate regional warming during summer and the rainy season.

In comparison, DTR shows a consistent decline. The Northern region experiences precipitation trends in February, May, and November. T_{mean} reveals marked warming in March and August, while DTR continues to decline. Lastly, the Southern region exhibits notable increases in precipitation, with strong seasonal trends in winter and the rainy season. T_{mean} increases in several months, while DTR declines in January. These findings underscore heightened precipitation and warming across Thailand, with complex DTR variability.

The trend analysis of precipitation, T_{mean} , and DTR across all regions reveals key regional variations while aligning with broader CC patterns. The rainy season shows a general increase in precipitation in the Central and Eastern regions. However, the Northeast exhibits mixed results, with decreases in June but increases in July and September, highlighting high intra-seasonal variability. These variations suggest that local factors, such as topography, large-scale teleconnections, and regional weather systems, shape precipitation trends. T_{mean} across all regions indicates consistent warming during the summer and rainy seasons. This warming reflects the influence of global climate change and is most pronounced in regions like the North and East. The declining trend in DTR during the winter and rainy seasons indicates a reduction in day-night temperature differences. This pattern is most noticeable in the Eastern and Central regions, where reduced temperature variability could affect water management practices. The findings indicate a coherent pattern of increased rainfall during monsoon months, though regional differences persist. The Northeast stands out for its more erratic

precipitation trends, reflecting local climate complexities. The T_{mean} warming and decreasing DTR trends suggest that global CC exerts pressure on Thailand's local climate systems, necessitating adaptive strategies in agriculture, water management, and climate resilience. The analysis underscores the need for region-specific interventions to mitigate the diverse impacts of CC across Thailand.

3.1.6. Forecasts of precipitation, T_{mean} , and DTR

The study offers new insights into forecasting climate variability based on precipitation, T_{mean} , and DTR between 2023 and 2038 using two DL algorithms: Wavelet-CNN-LSTM and Wavelet-LSTM. The comparison between the Wavelet-LSTM and Wavelet-CNN-LSTM models across various climate regions shows distinct accuracy and error metrics patterns, as shown in Table 4.

For DTR, the Wavelet-CNN-LSTM performs better than the Wavelet-LSTM. The higher R^2 (0.73–0.85) indicates a better model fit in the northern and southern regions. The lower RMSE and RRMSE for the Wavelet-CNN-LSTM suggest reduced error. In contrast, the lower MAPE indicates improved forecasting accuracy, especially in the Northern and Southern regions, and the same trends were observed in the central and eastern regions. In the northeastern region, Wavelet LSTM performed better than Wavelet-CNN-LSTM. In precipitation forecasting, the results are mixed. Although the Wavelet-LSTM shows slightly better performance in all regions except central, the overall differences in R^2 , RMSE, and MAE between the models are minor. Both models struggle more with precipitation, particularly in the Southern region, as reflected in the low R^2 , high RMSE, and RRMSE, indicating challenges in capturing precipitation variability. For T_{mean} , both models perform similarly with high R^2 (0.54–0.87) across all regions, suggesting vital model accuracy. The Wavelet-CNN-LSTM performs marginally better in some regions, particularly in minimizing errors (lower RMSE and MAPE), notably in the Southern region. The forecasts for DTR, precipitation, and T_{mean} for years 2023–2028 are provided in the Supplementary file. Wavelet-CNN-LSTM tends to provide more consistent performance, especially for DTR and T_{mean} forecasting.

The descriptive statistics of the forecasted climate variables for each region are detailed in Figs. 4–8. These figures illustrate the mean and standard deviation (SD) for historical (1993–2022) and forecasted trends. For instance, Fig. 4(a) presents the historical precipitation data for the Central region, with a mean of 66.74 mm and an SD of 38.84 mm. In contrast, the forecasted precipitation using the Wavelet-LSTM model

Table 4
Forecasting models performance evaluation.

Metrics Models	Central Region		Eastern Region		Northeastern Region		Northern Region		Southern Region	
	Wavelet-LSTM	Wavelet-CNN-LSTM	Wavelet-LSTM	Wavelet-CNN-LSTM	Wavelet-LSTM	Wavelet-CNN-LSTM	Wavelet-LSTM	Wavelet-CNN-LSTM	Wavelet-LSTM	Wavelet-CNN-LSTM
DTR										
R^2	0.52	0.67	0.61	0.65	0.80	0.73	0.71	0.73	0.32	0.85
RMSE	1.07	0.88	1.17	1.11	1.25	1.45	2.03	1.95	4.01	1.08
RRMSE (%)	19.37	15.35	17.61	16.22	13.32	17.23	15.12	14.79	21.37	21.7
MAE	0.88	0.68	0.9	0.88	1.04	1.12	1.53	1.42	3.20	1.2
MAPE (%)	7.55	6.36	8.71	8.46	8.44	8.84	11.13	9.88	19.93	9.00
Precipitation										
R^2	0.72	0.63	0.80	0.82	0.58	0.60	0.62	0.65	0.24	0.32
RMSE	26.00	29.79	20.91	20.34	57.54	56.23	56.32	53.75	92.86	93.61
RRMSE (%)	14.5	16.88	14.86	14.35	16.08	15.41	17.21	18.02	31.02	23.94
MAE	20.3	23.05	15.8	15.86	42.55	41.8	38.32	40.03	73.91	73.37
MAPE (%)	33.72	39.39	22.91	22.52	44.23	35.51	40.01	39.34	36.09	32.36
T_{mean}										
R^2	0.66	0.69	0.64	0.54	0.82	0.82	0.87	0.86	0.57	0.72
RMSE	1.32	0.02	1.13	1.27	1.34	1.35	1.19	1.20	0.97	0.79
RRMSE (%)	19.51	17.32	20.86	20.44	13.88	15.05	12.24	12.12	14.36	11.82
MAE	1.04	0.93	0.90	0.90	1.05	1.03	0.91	0.91	0.63	0.53
MAPE (%)	3.44	8.68	2.99	3.31	3.57	3.54	3.14	3.09	2.13	1.82

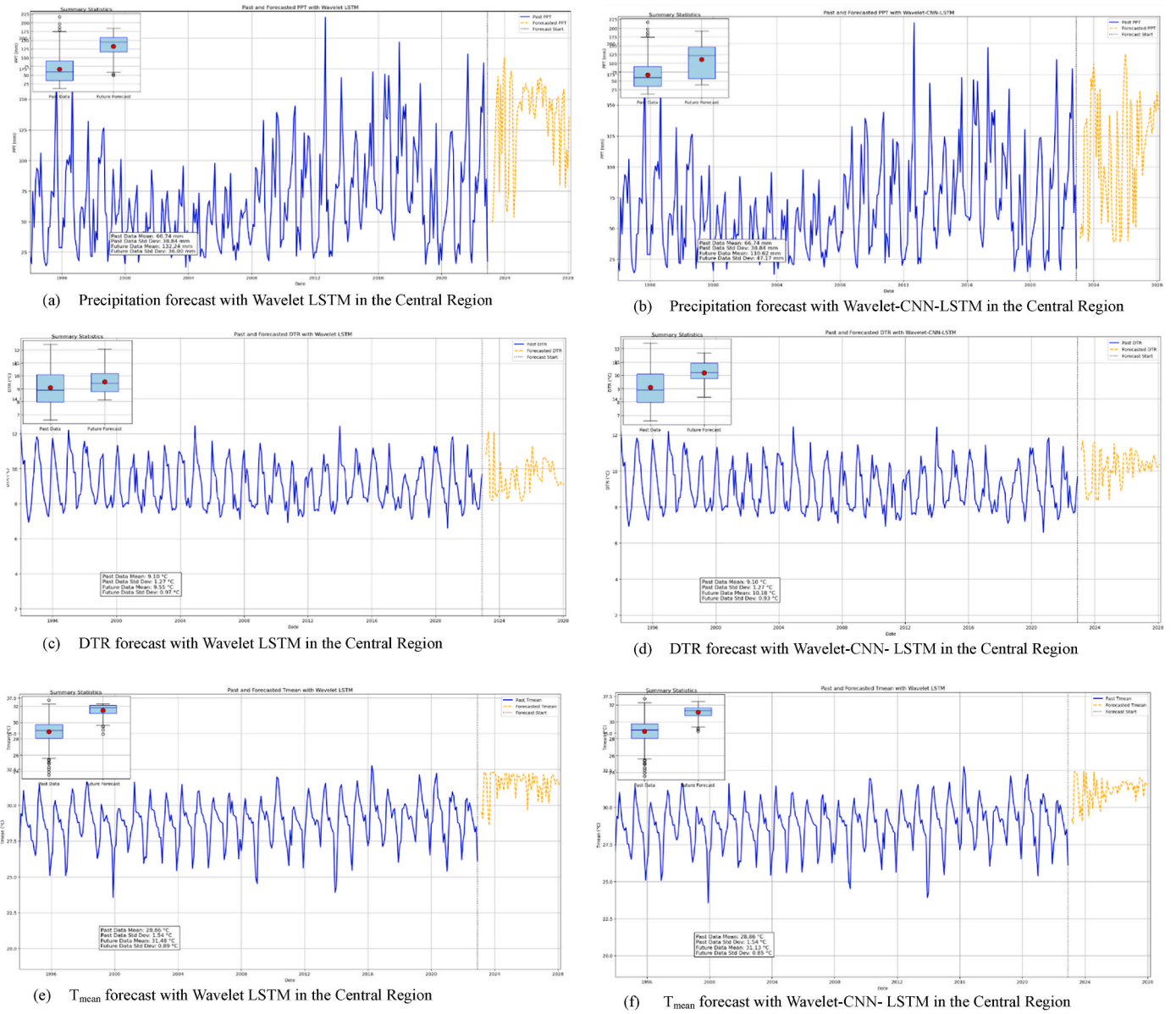


Fig. 4. Forecasting Results of precipitation, T_{mean} , and DTR with both DL models for the Central Region.

has a mean of 132.24 mm and an SD of 36.00 mm, while the Wavelet-CNN-LSTM model predicts a mean of 110.62 mm and an SD of 47.62 mm. It suggests that although the Wavelet-CNN-LSTM model forecasts composite average precipitation, its predictions are more aligned with the statistical distribution of the historical data, albeit indicating an increase in future precipitation for the Central region.

Similar patterns are observed across other figures, which compare past and forecasted trends in T_{mean} , precipitation, and DTR for all five regions of Thailand. These comparisons provide insights into regional climate variability and highlight the potential shifts in key climate variables, emphasizing the models' ability to capture both the spatial and temporal dynamics of future climate scenarios.

The Wavelet-CNN-LSTM model outperforms in forecasting DTR and T_{mean} in regions such as the North and South, where climatic variability is less pronounced. This advantage arises from the CNN layer's ability to capture spatial features, such as temperature gradients, which enhances performance when combined with LSTM's capacity to model temporal dependencies. The incorporation of Wavelet decomposition further improves both models by identifying high-frequency temporal patterns, enabling more accurate trend detection in temperature-related data.

However, in precipitation forecasting, the Wavelet-CNN-LSTM model shows superior performance compared to Wavelet-LSTM in the North-eastern region, likely due to its stronger focus on capturing temporal dependencies, which are crucial for modeling precipitation patterns. Precipitation's inherent non-linearity and stochastic nature, particularly in the South, present challenges for both models. Although the Wavelet-CNN-LSTM model captures spatial variations, it may not fully account for the localized nature of rainfall events, suggesting the need for further refinements, such as region-specific tuning. These regional discrepancies underscore the importance of calibrating models based on the unique climatic characteristics of each region to enhance predictive accuracy. Additional atmospheric variables may benefit improved precipitation forecasts, particularly in regions with complex weather dynamics, where local microclimates and topography hinder model performance.

The forecasting results for 2023–2028 require further elaboration to capture the uncertainty inherent in climate predictions. Incorporating prediction or confidence intervals for precipitation forecasts would provide a clearer understanding of the variability in projected trends. For example, a 95 % confidence interval would offer stakeholders insight into the expected range of climate outcomes in regions with

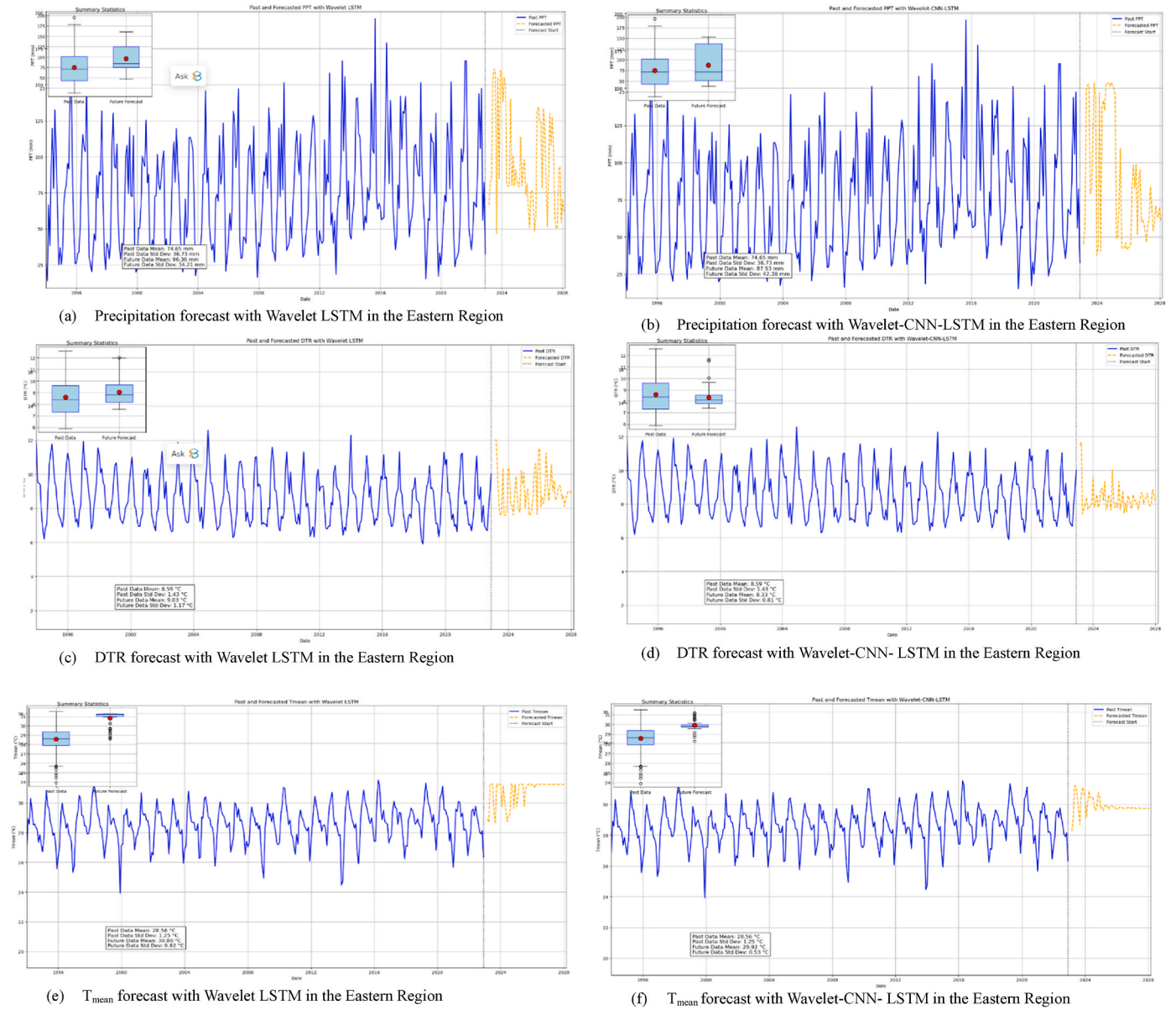


Fig. 5. Forecasting Results of precipitation, T_{mean} , and DTR with both DL models for the Eastern Region.

climate variability. Scenario-based forecasting and ensemble modeling techniques could further strengthen the robustness of these projections, enabling a more comprehensive understanding of future climate dynamics.

4. Discussion

The analysis of climate trends in Thailand reveals diverse regional patterns, as detailed in Tables 1–3 and S1 to S12. In the Central region, there is an increase in precipitation, a trend consistent with the research by Limsakul (2021), which documented increased precipitation during the monsoon season in Thailand [83]. The observed trends in the rainy season align with the broader findings of Faikruea et al. (2020), who noted increased rainfall over the years [84]. However, this study identifies a more pronounced trend in August and September than the research, indicating regional variability within the Central region.

T_{mean} data for the Central region show warming in March, August, and September. This finding corroborates the results of Manton et al. (2001), who reported warming across Southeast Asia, specifically in Thailand [85]. Nonetheless, the degree of warming observed here is

somewhat higher than the national averages Limsakul (2020) reported, suggesting a potentially more acute warming trend in the Central region [84]. The DTR analysis reveals a general decline in DTR, consistent with the findings of Cheong et al. (2018), who observed a decline across Southeast Asia [86]. However, the magnitude of this decline is less pronounced compared to other regional studies, such as Imran et al. (2023), which documented more substantial reductions in DTR in Asia [87]. In the Eastern region, increases in precipitation are observed, aligning with Deb et al. (2018), who also reported increased rainfall during these months [88]. These trends are consistent with the results of Nounmusig (2018), which indicated similar increases in precipitation during the monsoon period [49]. T_{mean} trends in the Eastern region reveal warming in March, May, June, August, September, October, and November. These findings align with Vongvisessomjai (2010), who observed warming trends in Eastern Thailand [89]. The warming trends are consistent with Kachenchart et al. (2021), who reported similar patterns across the region, suggesting that the observed temperature increases are aligned with broader climatic changes in Thailand [90]. DTR analysis in the Eastern region indicates a general decline with negative trends. The extent of this decline is comparable to the broader

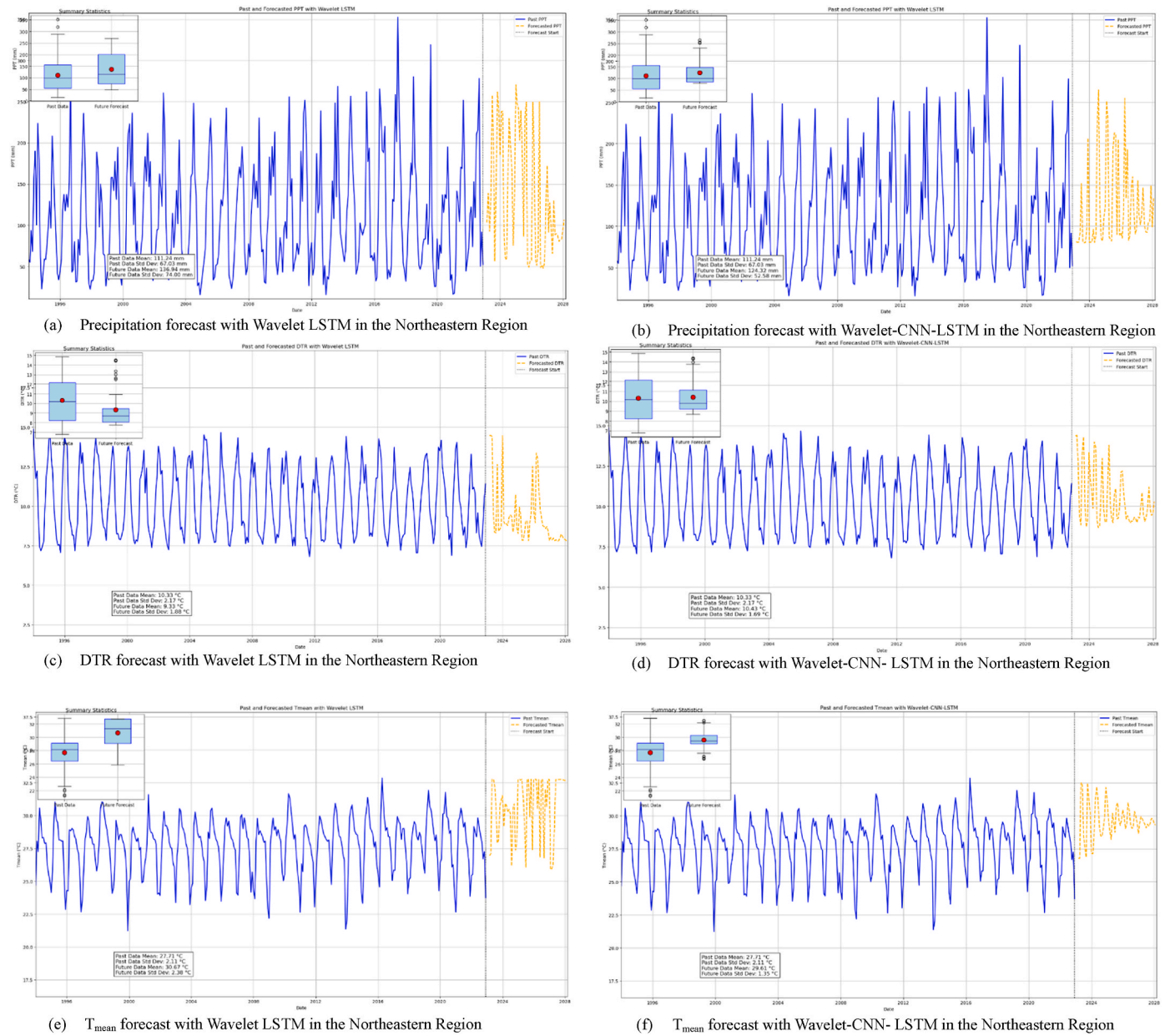


Fig. 6. Forecasting Results of precipitation, T_{mean} , and DTR with both DL models for the Northeastern Region.

regional trends reported by Cheong et al. (2018), indicating that the Eastern region follows similar patterns [86]. In the Northeastern region, precipitation trends show increases in May, July, and September, contrasted with a decrease in June. These results reflect a more complex precipitation pattern than Masud et al. (2016), which reported less evident variability [38]. The Northern region exhibits increases in precipitation in February, May, and November, with notable rises during these months. These findings align with Moazzam et al. (2020), who documented similar trends in Northern Thailand [25]. Temperature trends in the Northern region show warming across various months, consistent with the findings of Gunathilake et al. (2020) and Humphries et al. (2024), who reported warming trends in Northern Thailand [28, 91]. The observed trends in both precipitation and temperature suggest a general alignment with broader regional findings. In the Southern region, increases in precipitation are observed in February, May, and June, with a pronounced rise in February (Sen's slope of 3.361 mm/year). These trends reflect ongoing climatic shifts consistent with Kamworapan and Surussavadee (2019), who reported similar increases

in rainfall [92]. The warming trends observed in several months also align with broader regional observations, supporting the findings of Humphries et al. (2024a) regarding increased temperatures in the region [28]. Overall, the observed trends in precipitation, temperature, and DTR across various regions in Thailand highlight regional and broader climatic changes. The results are consistent with findings from other studies, which all stated the shifts in precipitation patterns and temperature increases, with varying impacts on DTR variability.

In a comparison of these findings with the literature, DTR forecasting, models' R^2 (0.73–0.85) aligns with the findings of Banadkooki et al. (2019), which showed that LSTM models achieved R^2 between 0.60 and 0.85 when forecasting temperature-related parameters in complex climatic conditions [93]. The lower performance observed in the southern region ($R^2 = 0.32$) is attributed to higher climatic variability, a challenge similarly noted by Sha et al. (2020), where complex terrain and microclimates led to decreased model accuracy [94]. R^2 for precipitation forecasting (0.24–0.82) also falls within a range observed in comparable studies using DL models. Vivas et al. (2022) reported R^2

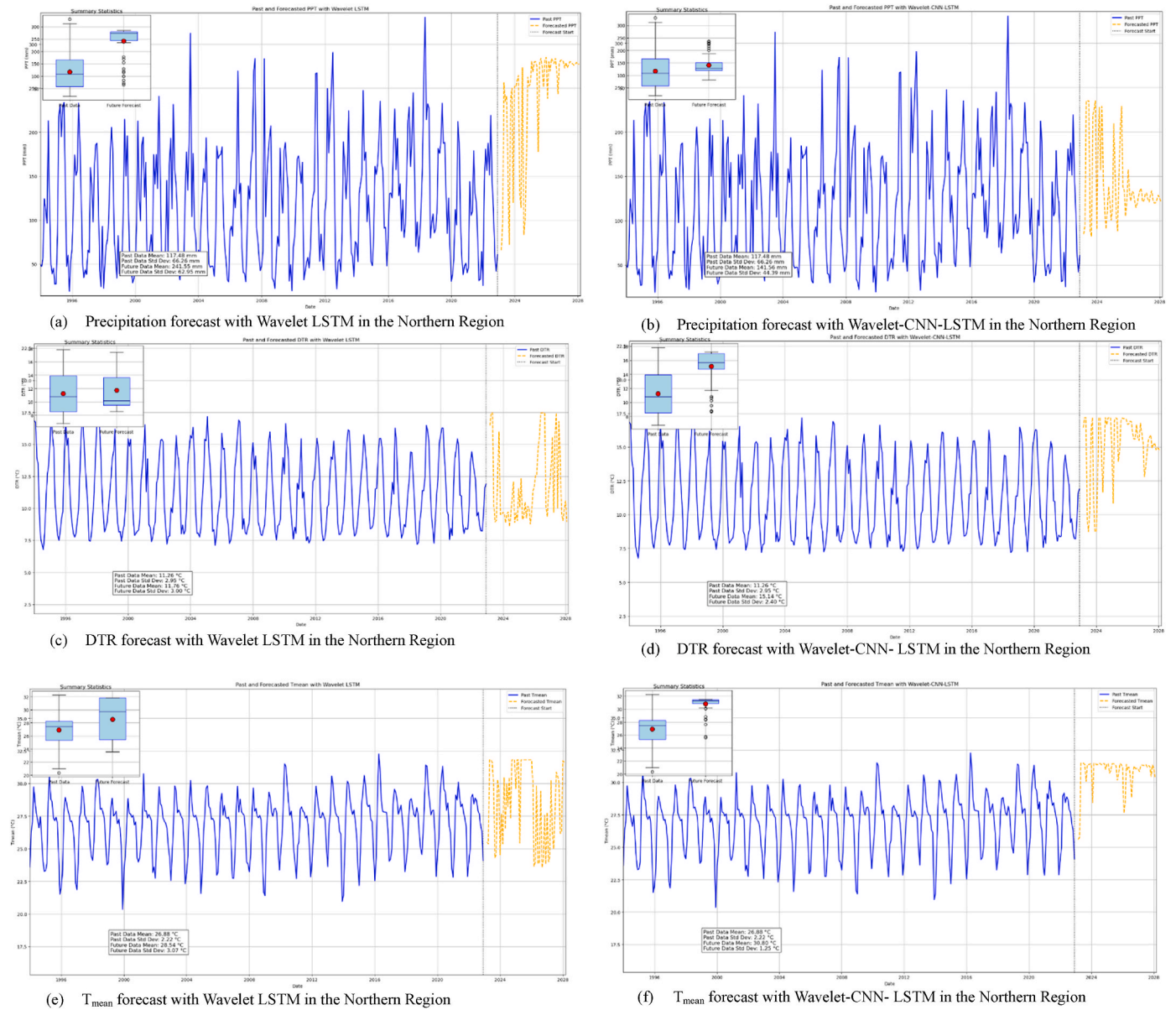


Fig. 7. Forecasting Results of precipitation, T_{mean} , and DTR with both DL models for the Northern Region.

from 0.30 to 0.78 in precipitation forecasting using a deep wavelet-lagged approach across tropical regions [95]. Higher MAE results in southern, northern, and northeastern regions may be because model performance often declines in areas with high precipitation variability due to rainfall patterns' non-linear and stochastic nature [47]. It suggests that both models perform well under certain conditions but face challenges in regions with more complex and variable precipitation dynamics [96].

For T_{mean} , both models demonstrated relatively strong performance, with R^2 (0.82–0.86), in the northern region. These results are comparable to those obtained by Hou et al. (2022), where CNN-LSTM models achieved $R^2 = 0.70$ to 0.90 in temperature forecasting across various regions in China [97]. The lower R^2 in some regions (e.g., central and southern regions) is due to regional climatic complexities, which was also observed by Chapagain et al. (2020) in temperature forecasting over Thailand [98]. These variations underscore the impact of local climatic factors on model accuracy when dealing with temperature data, which exhibit more consistent patterns than precipitation. DTR and T_{mean} exhibit more consistent patterns, allowing the models to capture trends more effectively, reflected in higher R^2 . Precipitation, however, is

influenced by numerous factors, such as local convective systems, seasonal monsoons, and geographical features, leading to higher model uncertainty. These findings align with the study by Muhammad et al. (2018), where region-specific model calibration improved model performance in precipitation forecasting [99]. The use of Wavelet decomposition in conjunction with LSTM and CNN-LSTM in the study is a well-established approach for improving model performance by capturing temporal and frequency-domain data features [39]. However, the variability in model performance across regions suggests that further refinement, such as region-specific tuning or incorporating additional climatic variables, could enhance predictive accuracy, particularly for precipitation forecasting.

5. Conclusion

The analysis of climate variability, trends, and forecasting for Thailand provides valuable insights into the dynamics of regional climate behavior. The study revealed regional differences in precipitation, T_{mean} , and DTR trends, critical for informed water resource management and planning in a climate-sensitive country like Thailand. The

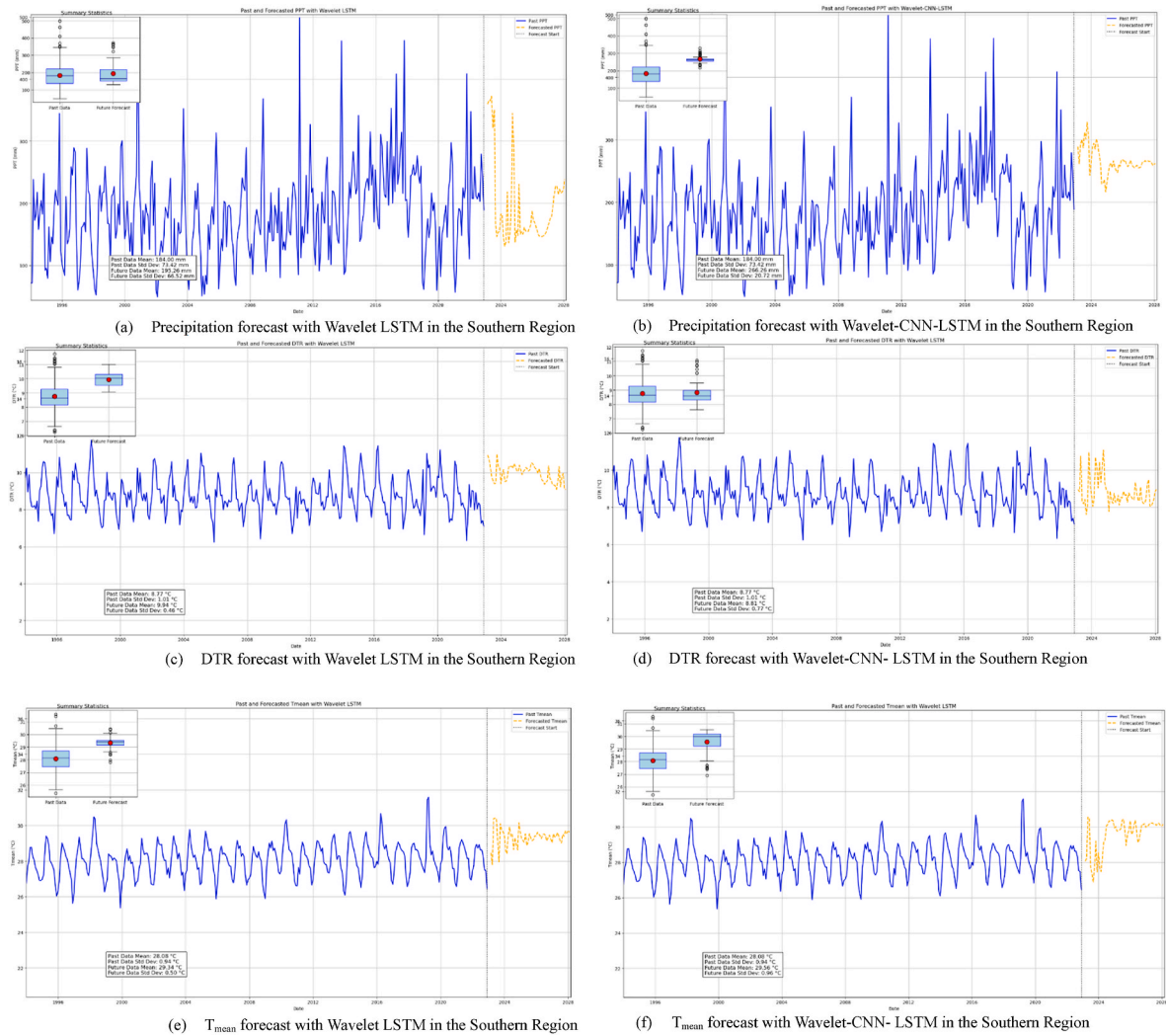


Fig. 8. Forecasting Results of precipitation, T_{mean} , and DTR with both DL models for the Southern Region.

Central and Southern regions experienced notable increases in precipitation and temperature, with annual rainfall rising by 0.093 mm/year and 0.148 mm/year, respectively, and mean temperatures increasing by 0.002 °C/year in both regions. Similar trends were observed in the Eastern region. In contrast, the Northeastern region exhibited more complex patterns, with increases in rainfall during specific months but modest overall annual trends. A decrease in DTR across all regions suggests reduced daily temperature variability.

These trend analyses highlight climate change's varying impacts on precipitation and temperature at the regional scale dataset. It also highlights the importance of studying these variables to understand the impact of climate change on any region. The study presented a novel hybrid DL wavelet decomposed CNN LSTM model to improve prediction accuracy for forecasting climate variables. The proposed model's predictive accuracy is better than wavelet-LSTM, particularly in the Northern region, where R^2 up to 0.87 for T_{mean} and DTR from 2023 to 2028. The model exhibited more substantial consistency across regions for DTR and T_{mean} forecasting. However, both models faced challenges with precipitation forecast, particularly in the Central and Northeastern regions, emphasizing the need for more influential climate variables to regionally capture the precipitation patterns and then need a refined model. Overall, this integration of wavelet decomposition with LSTM and CNN models offers a refined approach for capturing complex climate dynamics in precipitation and temperature, especially at a regional level dataset. The research contributes to improving climate

prediction methods, identifying regional challenges in forecasting precipitation, and stressing the importance of tailored models for specific areas. These advancements are crucial for supporting water resource management and mitigating the impacts of climate variability in Thailand. For future research to understand climate variability, it is recommended to include large-scale teleconnections such as El Niño-Southern Oscillation (ENSO), Madden-Julian oscillation (MJO), etc., in DL modeling.

List of Abbreviations

CC	Climate Change
DDMs	Data-Driven Methods
MDMs	Model-Driven Methods
ENSO	El Niño-Southern Oscillation
MJO	Madden-Julian oscillation
AI	Artificial Intelligence
MK	Mann-Kendall
SSE	Sen's slope estimator
DTR	Diurnal Temperature Range
GCM	Global Circulation Model
DL	Deep learning
ML	Machine learning
LSTM-RNN	Long Short-Term Memory Recurrent Neural Networks
SD	Standard Deviation
LSTM	Long short-term memory
SWAT	Soil and Water Assessment Tool

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(continued)

CNN	Convolutional Neural Network
TMD	Thai Meteorological Department
ANNs	Artificial Neural Network
FT	Fourier Transform
STFT	Short-Time Fourier Transform
WT	Wavelet Transformation
CWT	Continuous Wavelet Transform
DWT	Discrete Wavelet Transform
MLP	Multilayer Perceptron
ReLU	Rectified Linear Unit
R ²	Coefficient of Determination
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error

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CRedit authorship contribution statement

Muhammad Waqas: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Usa Wannasingha Humphries:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Phyo Thandar Hlaing:** Visualization, Validation, Software, Data curation.

Declaration of Competing Interest

The authors declare that the research was conducted without any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability

Data will be made available on request.

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